METFORMIN

Comprehensive Drug Monograph & Clinical Practice Guidelines

BOXED WARNING: LACTIC ACIDOSIS

Post-marketing cases of metformin-associated lactic acidosis have resulted in death, hypothermia, hypotension, and resistant bradyarrhythmias. The onset is often subtle, accompanied only by nonspecific symptoms such as malaise, myalgias, respiratory distress, somnolence, and abdominal distress. Risk factors include renal impairment, concomitant use of certain drugs, age ≥ 65 years, a radiological study with contrast, surgery, hypoxic states, and hepatic impairment. If metformin-associated lactic acidosis is suspected, immediately discontinue the drug and institute general supportive measures in a hospital setting. Prompt hemodialysis is recommended.

1. General Information

- **Generic Name:** Metformin hydrochloride
- **Therapeutic Class:** Biguanide Antidiabetic Agent
- **Chemical Name:** 1,1-Dimethylbiguanide hydrochloride
- **Molecular Formula:** $C_4H_{11}N_5 \cdot HCl$

2. Clinical Pharmacology

Mechanism of Action

Metformin is an antihyperglycemic agent which improves glucose tolerance in patients with type 2 diabetes mellitus (T2DM), lowering both basal and postprandial plasma glucose. It does not stimulate insulin secretion and therefore does not produce hypoglycemia when used as monotherapy. Metformin acts via three main mechanisms:

1. Reduction of hepatic glucose production by inhibiting gluconeogenesis and glycogenolysis.
2. In muscle, it increases insulin sensitivity, improving peripheral glucose uptake and utilization.
3. Delay of intestinal glucose absorption.

At the molecular level, metformin activates adenosine monophosphate-activated protein kinase ($AMPK$), a crucial cellular energy sensor, which downregulates lipogenesis and suppresses key gluconeogenic enzymes.

Pharmacokinetics

- **Absorption:** Absolute bioavailability of an oral 500 mg dose is approximately 50% to 60%. Food decreases the extent and slightly delays the absorption.
- **Distribution:** Metformin is negligibly bound to plasma proteins; it partitions into erythrocytes. Steady-state is reached within 24–48 hours.
- **Metabolism:** Intravenous single-dose studies demonstrate that metformin is excreted unchanged in the urine and does not undergo hepatic metabolism or biliary excretion.

- **Elimination:** Renal elimination via glomerular filtration and active tubular secretion. The plasma elimination half-life is approximately 6.2 hours. In blood, the elimination half-life is approximately 17.6 hours.

3. Indications & Clinical Usage

- **Type 2 Diabetes Mellitus:** Indicated as an adjunct to diet and exercise to improve glycemic control in adults and pediatric patients aged 10 years and older.
- **Off-Label Uses:** Prediabetes management, Polycystic Ovary Syndrome (PCOS) insulin-resistance, and antipsychotic-induced weight gain attenuation.

4. Dosage and Administration

Adult Dosage (Immediate-Release)

- **Starting Dose:** 500 mg orally twice daily or 850 mg once daily, given with meals.
- **Titration:** Increase in increments of 500 mg weekly or 850 mg every 2 weeks, up to a maximum dose of 2,550 mg/day given in divided doses.
- **Extended-Release (XR):** Start with 500 mg to 1,000 mg once daily with the evening meal. Titrate by 500 mg weekly to a maximum of 2,000 mg once daily.

Pediatric Dosage (≥ 10 years of age)

- **Starting Dose:** 500 mg twice daily.
- **Titration:** Increase by 500 mg weekly; maximum recommended dose is 2,000 mg/day in divided doses. (Extended-release forms are not approved for pediatrics).

5. Contraindications

1. Severe renal impairment (eGFR below 30 mL/min/1.73 m^2).
2. Known hypersensitivity to metformin hydrochloride.
3. Acute or chronic metabolic acidosis, including diabetic ketoacidosis, with or without coma.

6. Warnings, Precautions & Drug Interactions

Renal Impairment Guidelines

Assess renal function prior to initiation and periodically thereafter. Dosing recommendations are strongly dictated by the Estimated Glomerular Filtration Rate (eGFR):

eGFR Level (mL/min/1.73 m^2)	Clinical Action / Recommendation
≥ 45	No dosage adjustment necessary. Monitor renal function annually.
30 to 44	Initiation is not recommended. For patients already taking metformin, consider a 50% dose reduction and monitor eGFR every 3 months.
< 30	Contraindicated. Discontinue immediately.

Iodinated Contrast Imaging Procedures

Discontinue metformin at the time of, or prior to, an iodinated contrast imaging procedure in patients with an eGFR between 30 and 60 mL/min/1.73 m^2, in patients with a history of hepatic impairment, alcoholism, or heart failure, or in patients who will receive intra-arterial contrast. Re-evaluate eGFR 48 hours after the procedure; restart if renal function is stable.

Key Drug Interactions

- **Carbonic Anhydrase Inhibitors (e.g., Topiramate, Zonisamide):** May increase the risk of lactic acidosis.
- **Cationic Drugs / OCT2 Inhibitors (e.g., Dolutegravir, Cimetidine, Ranolazine):** May increase metformin systemic exposure by competing for common renal tubular transport systems.
- **Glucocorticoids, Thiazides, and Other Diuretics:** Tend to produce hyperglycemia and may lead to loss of glycemic control.

7. Adverse Reactions

The most common adverse reactions are gastrointestinal in nature, usually resolving spontaneously over time or minimized by dose titration and administration with meals:

- Diarrhea (53%), Nausea/Vomiting (26%), Flatulence (12%), Abdominal discomfort (6%).
- **Vitamin B12 Deficiency:** Long-term metformin use has been associated with a decrease in serum vitamin B12 levels. Periodic monitoring of hematologic parameters (e.g., hemoglobin/hematocrit and B12 levels) is advised every 2 to 3 years.

8. Clinical Guidelines Summary

According to current major clinical guidelines (such as the American Diabetes Association [ADA] Standards of Care):

- **First-line Therapy:** Metformin remains the foundational preferred initial pharmacologic agent for the treatment of type 2 diabetes, unless explicit contraindications exist.
- **Combination Strategy:** If the target HbA1c is not achieved within 3 months, metformin should be continued in combination with other agents (e.g., SGLT2 inhibitors, GLP-1 receptor agonists, DPP-4 inhibitors, or basal insulin) individualized to patient comorbidities (e.g., atherosclerotic cardiovascular disease, heart failure, chronic kidney disease).
- **Cardiovascular/Renal Protection:** In patients with established ASCVD, heart failure, or CKD, second-line therapies with proven benefit (SGLT2 inhibitors or GLP-1 RAs) should be co-prescribed early, independent of baseline HbA1c levels or the metformin dose.